

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-blind, Active Comparator-controlled Clinical Study to Evaluate the Safety, Tolerability, and Immunogenicity of V116 in Pneumococcal Vaccine-naïve Adults 50 Years of Age or Older **Short Title/Acronym:** Safety and Immunogenicity of V116 in Pneumococcal Vaccine-naïve Adults 50 Years of Age or Older (STRIDE-10 / V116-010) **Protocol Number:** V116-010-01 **NCT Number:** NCT05569954 **Sponsor Name:** Merck Sharp & Dohme LLC (MSD) **Investigational Product:** V116 (CAPVAXIVE™, 21-valent pneumococcal conjugate vaccine) **Phase of Development:** Phase 3 **Medical Monitor:** MSD Clinical Trial Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the safety, tolerability, and immunogenicity of V116 compared to the active comparator (PPSV23) in pneumococcal vaccine-naïve adults ≥ 50 years of age. The study aims to demonstrate that at 30 days post-vaccination, the immunogenicity of V116 is noninferior to PPSV23 for 12 common serotypes, and superior for the 9 serotypes unique to V116. * **Secondary Objectives:** To assess the percentage of participants with solicited injection-site and systemic adverse events (AEs), and to determine whether V116 is superior to PPSV23 based on the proportion of participants achieving a ≥ 4 -fold rise from baseline in unique V116 serotypes as measured by serotype-specific opsonophagocytic activity (OPA) geometric mean titers (GMTs).

2.2 Background & Rationale Streptococcus pneumoniae is a major cause of vaccine-preventable disease worldwide. Despite the availability of pediatric vaccination programs, adult invasive pneumococcal disease (IPD) remains a significant unmet medical need. V116 is specifically designed for adults to provide broader coverage against the specific serotypes responsible for the majority of residual adult IPD cases.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator-controlled (PPSV23 / PNEUMOVAX™ 23) * **Blinding:** Double-blind * **Randomization:** 1:1 ratio

3.2 Planned Enrollment & Duration * **Target N:** Approximately 1,400 evaluable participants * **Number of Sites:** 55 locations across 11 countries (e.g., Argentina, Australia, Colombia,

Germany, Israel, New Zealand, South Korea, Spain, etc.) *
Estimated Study Start: 06-Nov-2022 * **Estimated Primary Completion:** 30-Oct-2023 * **Study Completion (Actual):** 07-Feb-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants ≥ 50 years of age. 2. Participants who are pneumococcal vaccine-naïve (have not previously received any pneumococcal vaccine).

4.2 Exclusion Criteria 1. Received prior pneumococcal vaccine or expected to receive any pneumococcal vaccine during the study outside the protocol. 2. Known malignancy that is progressing or has required active treatment < 3 years before enrollment. 3. Recent febrile illness (temperature $\geq 100.4^{\circ}\text{F}/38.0^{\circ}\text{C}$) or received antibiotic therapy for any acute illness occurring < 72 hours before receipt of study vaccine. 4. Has a coagulation disorder contraindicating intramuscular (IM) vaccination.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Single dose (0.5 mL) of V116 or PPSV23 administered on Day 1. * **Route of Administration:** Intramuscular (IM) injection. * **Duration of Treatment:** Single administration with a follow-up period spanning 6 to 24 months.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard daily medications not interfering with the immune response. * **Prohibited:** Administration of any non-study pneumococcal vaccines during the trial; participating in another interventional clinical study within 6 months prior (specifically per South Korea country requirements).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening / Baseline	Day 1	Informed Consent, Medical History, Randomization, Blood Draw (Baseline Immunogenicity), Single IM Vaccination
Interim 1	Days 1 - 5	Participant electronic diary for solicited injection-site and systemic AEs
Interim 2	Day 30	Safety review, Blood Draw for post-vaccination Immunogenicity assessment (OPA GMTs / PnECL)
End of Study	Month 6 (or up to Month 24)	Safety follow-up (up to 6 months post-vaccination for vaccine-related SAEs); optional extensions for PBMC substudies

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Serotype-specific Opsonophagocytic Activity (OPA) Geometric Mean Titers (GMTs) at 30 days post-vaccination; Percentage of participants with solicited injection-site and systemic AEs up to Day 5 post-vaccination. * **Measurement Method:** Central Laboratory Analysis utilizing Pneumococcal Electrochemiluminescence (PnECL) and OPA assays.

7.2 Secondary & Exploratory Endpoints * Proportion of participants with a ≥ 4 -fold rise from baseline to 30 days post-vaccination in OPA responses for unique V116 serotypes. * Serotype-specific Immunoglobulin G (IgG) Geometric Mean Concentrations (GMCs).

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Patient diary collection for solicited injection-site AEs (pain/tenderness, redness/erythema, swelling) and systemic AEs (fatigue, headache, myalgia, pyrexia) from Day 1 to Day 5.
 - **Serious Adverse Events (SAEs):** Routine recording of any vaccine-related SAEs up to 6 months post-vaccination.
 - **Data Monitoring Committee (DMC):** Utilized standard Sponsor-directed (MSD) safety monitoring oversight.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Per-Protocol (PP) set for Immunogenicity outcomes; Safety Set for AE/SAE assessment.
 - **Sample Size Justification:** $\sim 1,400$ patients powered to determine noninferiority for 12 common serotypes and superiority for 9 unique serotypes based on pre-defined margins.
 - **Handling of Missing Data:** The GMC/GMT ratio estimation and 95% CIs were calculated using a Constrained Longitudinal Data Analysis (cLDA) method; Miettinen & Nurminen method applied for between-group differences.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP), the Declaration of Helsinki, FDAAA, and the EMA Clinical Trials Regulation 536/2014.
- **Monitoring Plan:** Scheduled onsite and remote source data verifications by Merck Sharp & Dohme LLC.

- **Audit Trail:** Assured through robust EDC (electronic data capture) systems logging all query resolutions and eCRF locking.
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11. Study Identifiers & Governance

- **Primary ID:** {{V116-010}}
 - **Registry Number:** {{NCT05569954}}
 - **Sponsor/Lead Organization:** {{Merck Sharp & Dohme LLC}}
 - **Study Title:** {{STRIDE-10}}
 - **Official Regulatory Title:** {{A Phase 3, Randomized, Double-blind, Active Comparator-controlled Clinical Study to Evaluate the Safety, Tolerability, and Immunogenicity of V116 in Pneumococcal Vaccine-naïve Adults 50 Years of Age or Older}}
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12. Clinical Framework (Pneumococcal Specific)

- **Primary Condition:** {{Pneumococcal Disease}}
 - **Diagnosis Threshold:** {{Pneumococcal Vaccine-naïve Adults \$ge 50\$ Years of Age}}
 - **Medical Methodology:** {{Active Immunization}}
 - **Optimization Target:** {{Prevention of invasive pneumococcal disease and pneumonia}}
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13. Study Procedures & Methodology

- **Management Pathway:** {{Single Intramuscular Vaccination}}
 - **Education Protocol (HCP):** {{Protocol training and AE reporting guidelines}}
 - **Education Protocol (Patient):** {{Informed consent and diary card training for AEs}}
 - **Quality Audit Mechanism:** {{Site monitoring and source data verification}}
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14. Observation & Follow-Up Schedule

- **Total Duration:** {{6 to 24 follow-up months}}
 - **Onsite Visit Cadence:** {{2 to 7 visits (Day 1, Day 30, Month 6, etc.)}}
 - **Remote Monitoring Frequency:** {{As needed per site SOPs}}
 - **Checklist Review Interval:** {{Post-vaccination Day 5 for solicited AEs}}
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15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					